

the devices decide how often to communicate with each other.

- Better encryption management facilities.
- Improved strength of security in paired connections.
- Near field communication cooperation. NFC is another wireless technology with a very small range of around 10 centimetres or 4 inches. It allows for the creation of automatic secure Bluetooth connections when NFC radio interface is available. An example is the automatic uploading of photos from a mobile phone to another device just by bringing it close to the electronic device.

Bluetooth 3.0 is the newest specification adopted by the Bluetooth SIG (Special Interest Group) on April 21st 2009. The main feature is AMP (Alternate MAC/PHY), the addition of 802.11 as a high speed transport system. UWB (ultra-wideband) is missing from the specification. 802.11 is associated with Wi-Fi and alternate MAC/PHY or AMP allows the use of alternate MAC and PHYs for transporting Bluetooth profile data. The conventional Bluetooth radio is still used as the primary device for device discovery, initial connection and profile configuration, but the AMP option is used for high speed data transfer, especially when a great deal of data needs to be sent. 3.0 also allows for faster connections for small data by passing few protocols and for more standardised encryption key sizes.

Bluetooth saves power effectively, and in turn improves efficiency

The newest rage in Bluetooth technology is the Bluetooth low energy technology. This is an additional protocol stack added to the Bluetooth specification, and it is compatible with existing stacks. Earlier similar attempts, such as Wibree and Bluetooth ULP (ultra low power) have been outdated with the introduction of low energy. Bluetooth low energy technology is an open radio technology for really small devices. It addresses devices with very low battery capacity and is easily integrated with traditional Bluetooth. People are aware that this has great implementation opportunities for applications such as medical instrumentation. Imagine a small micro-circuit enabled with